

Week 4 — Fiat problems, early digital money, consensus, smart contracts, tokenization, Layer 2, CBDC

1) Recap of money — now with the blockchain angle

The lecture begins by revising the types of money:

- barter
- commodity money
- representative money
- fiat money

It also reminds us that USD moved from representative money to fiat money in 1971.

This recap is not random. Sir is setting up the question:

if fiat money depends on trust, what happens when trust breaks?

And if traditional digital money depends on banks, what happens when banks fail, censor, or invade privacy?

So Week 4 is really about:

1. why fiat and bank-based systems can fail,
2. why early digital currencies were not enough,
3. why blockchain became attractive.

2) Does fiat always work?

The lecture says fiat works on **trust in the government**, and since it is not backed by a valuable commodity, if trust becomes low or lost, there can be:

- sharp uncontrolled increase in prices,
- more printing/minting of money,
- money quickly becoming worthless,
- hyperinflation.

This is very important.

2.1 What is fiat money really depending on?

Fiat money is not valuable because the paper itself is valuable. It is valuable because people trust:

- the state,
- the legal system,
- the central bank,
- and the broader economy.

So fiat value is social, legal, and institutional.

2.2 What if trust breaks?

If people stop trusting the government's ability to manage the currency, then they may lose confidence in holding that money.

Once that happens:

- sellers demand more money for the same goods,
- prices rise faster,
- government may print even more,
- and the currency can collapse further.

That destructive loop is hyperinflation.

2.3 Why does sir ask “How do BTC and ETH maintain or enhance value?”

This is a “food for thought” question in the slide.

He wants you to compare fiat with crypto:

- fiat depends heavily on government trust,
- but crypto tries to create value through scarcity, protocol rules, decentralization, utility, and demand.

He is not giving the full answer here yet — just making you think about value foundations.

3) Hyperinflation of Germany 1923 — why this example matters

This is one of the biggest topics in Week 4. The lecture uses Germany after World War I as a real historical example of what can happen when fiat-like money loses stability.

The slides explain:

- Germany suspended gold-backed currency during WWI
- wanted to conserve gold
- financed war through borrowing instead of taxes
- expected to win and recover through reparations
- but lost the war
- had massive debt
- printed more money without economic backing
- was forced to pay reparations under the Treaty of Versailles
- later suffered industrial disruption in the Ruhr
- kept printing more banknotes
- by November 1923, 1 USD = 4,210,500,000,000 German marks.

Let's understand the story properly.

3.1 Before the collapse

At first, the mark weakened from 4.2 to 7.9 per USD between 1914 and 1918. The lecture even says this was “not so bad. Yet!”

This shows that early decline may look manageable.

3.2 After losing the war

Germany lost WWI, and then:

- already had huge debt,
- had fewer resources,
- and still needed to pay reparations.

That means financial pressure increased enormously.

3.3 Treaty of Versailles pressure

The Allies demanded reparations payable in valuable forms like gold, foreign currency, coal, ships, etc.

This is important because Germany could not just print paper marks and call that enough. It needed real value sources.

3.4 Printing without backing

The lecture explicitly says Germany printed more money without any economic resources to back it.

That is the core inflation lesson:

if you massively increase money supply without corresponding economic productivity or backing, prices can explode.

3.5 Ruhr occupation and passive resistance

French and Belgian troops occupied the Ruhr, Germany's main industrial region, in January 1923. Germans responded with passive resistance:

- workers stopped working
- factories shut down
- productivity fell
- tax collection fell
- but the government still had to pay workers
- so it printed even more banknotes.

This is a vicious cycle:

less production + more printed money = even worse currency collapse.

3.6 The final collapse

The lecture's graph and highlighted figure show that by November 1923, one USD was worth **4,210,500,000,000 German marks**. The chart on page 8 visually shows the exchange rate exploding upward almost vertically by 1923.

That number is the main memorization point.

3.7 Why the pictures matter

The slides include banknotes of:

- 50 billion marks,
 - 5 trillion marks,
 - 50 trillion marks,
- and photos of people handling money in absurd quantities. The images on pages 9–13 visually show how money became nearly worthless because denominations exploded.

These visuals are there to make the idea unforgettable:

when inflation becomes extreme, money stops functioning properly as money.

3.8 Human impact

The lecture includes two quotations:

- a man's long-term insurance payout ended up buying only a single loaf of bread,

- one US dollar was valuable enough to be treated like something worth setting up a trust around.

That makes hyperinflation personal. It destroys savings, planning, contracts, wages, and trust in the economic system.

3.9 The conceptual lesson

Sir is not teaching German history just for history's sake.

He is showing:

if money depends only on trust and policy, and that trust/policy collapses, the currency can fail catastrophically.

That helps explain why alternatives like crypto attract interest.

4) Digital currency and ePayments — what should digital money achieve?

The lecture then moves to digital money requirements. It says a good digital currency should provide:

- no stealing
- no double-spending
- minting / creation
- peer-to-peer payments
- no need for a third party to always be online
- no intermediary
- privacy at the level of printed currency.

This is one of the most important slide lists of Week 4.

4.1 No stealing

Only the rightful owner should be able to move the money.

4.2 No double-spending

The same unit of digital value should not be spendable twice.

4.3 Minting / creation

There must be a clear and fair rule for how new money is created.

4.4 Peer-to-peer payments

Alice should be able to pay Bob directly.

4.5 No always-online third party

The system should not depend on one bank server being constantly available.

4.6 No intermediary

The payment should not require a central middleman to approve it.

4.7 Privacy like printed cash

Traditional cash gives strong transaction privacy in normal use. A good digital equivalent should ideally preserve that to some degree.

This list is basically the checklist that older digital-money systems failed on — and that Bitcoin tried to solve.

5) Traditional ePayments

The lecture then studies the familiar bank-based digital payment model:

Alice wants to send \$5 to Bob, but the bank sits in the middle. The slides say the pros are that payment is digital and the users can be in different locations. But the cons are:

- not peer-to-peer
- bank can fail
- bank can delay or censor
- privacy issues because the bank knows who is paying whom and how much.

5.1 What is good about traditional ePayments?

They are convenient and digital. Two people do not need to meet physically. So in that sense they solve some problems of physical cash.

5.2 What is the main weakness?

They are not truly peer-to-peer.

Alice is not really paying Bob directly.
She is instructing the bank to update records.

5.3 Why is privacy weak?

Because the bank sees:

- sender,
- receiver,
- amount,
- timing,
- possibly purpose.

So the bank becomes a total visibility point.

5.4 Why can the bank fail?

The next slides emphasize exactly this. The lecture shows that banks can fail in reality, even using the example of ATM lines in Kabul after the US withdrawal in 08/2021.

So bank dependence is not only theoretical risk.

5.5 Why can the bank censor?

The lecture explicitly says banks can deny or censor customers, and mentions cases like Washington Mutual, Silicon Valley Bank, First Republic, and Signature Bank.

This is showing that centralized intermediaries can say “no,” either because of policy, failure, regulation, or their own bad business decisions.

So traditional ePayments solve distance and convenience, but not decentralization, censorship resistance, or privacy.

6) First attempt at P2P payments: Simple E-Cash

Now the lecture moves to early digital cash.

It shows a bank issuing a \$5 digital coin, but the problem is that a digital token can be copied just like a file. Alice can give the same token to Charlie later too. This is the **double-spending problem**.

6.1 Why is simple e-cash attractive?

Because it tries to make payment:

- digital
- peer-to-peer
- transferable between distant users

That is good.

6.2 What is the fatal problem?

A digital coin is just data.

And digital data can be copied.

So if Alice receives a digital coin file, what stops her from duplicating it and paying it again somewhere else?

This is the classic digital-money challenge.

6.3 Why is this different from physical cash?

Because physical notes cannot be effortlessly cloned like computer files.

That is why the transition from physical to digital money is harder than it first looks.

7) Simple e-cash with serial numbers

The next slide says the bank adds a **serial number (SN)**. Later, when the coin is spent, the serial number can be checked, and double-spending can be **detected**.

7.1 What does the serial number do?

It makes each coin unique.

So if the same serial number appears twice, the system knows the same coin has been spent more than once.

7.2 Does this solve double-spending fully?

Not really.

The lecture says simple e-cash can detect double-spending, but full P2P means it cannot always avoid it in advance. Avoiding it requires the bank to be online.

That is crucial.

So serial numbers help identify the problem, but they do not remove the need for a central checker.

7.3 Why is privacy damaged?

Because the bank knows the serial number, and therefore can connect the coin, payer, payee, and payment flow.

The lecture explicitly says privacy is violated because knowing SN lets the bank know who is paying how much to whom.

So simple e-cash improves digitization but fails on privacy and decentralization.

8) Simple E-Cash — pros and cons

The lecture summarizes:

Pros:

- digital
- payer and payee can be in different places
- peer-to-peer in spirit

Cons:

- double-spending can be detected but not fully prevented in full P2P mode
- avoiding it requires the bank to always be online
- the bank can fail, delay, or censor
- privacy is poor because the bank can identify transactions via serial numbers.

This is exactly the type of exam comparison question sir can ask.

9) Chaumian E-Cash using blinded signatures

Now comes a very important topic: **Chaumian e-cash**.

The lecture says:

- Alice chooses a serial number SN and blinds it

- she sends $b(SN)$ to the bank
- the bank signs $\text{sig}(b(SN))$
- bank does not see the real SN
- Alice can retrieve $\text{sig}(SN)$
- she gives $\text{sig}(SN)$, SN to Bob
- this solves the privacy issue of simple e-cash.

This is one of the smartest concepts in the course.

9.1 What is the big idea?

The big idea is:

get the bank to authorize the coin without letting the bank learn the actual serial number being authorized.

That is the privacy trick.

9.2 What is blinding?

Blinding means Alice mathematically hides the serial number before sending it to the bank.

So the bank signs a disguised version of the coin.

Later, Alice can remove the blinding effect and recover a valid signature on the real serial number.

9.3 Why is this brilliant?

Because now the bank has authorized the coin, but it does not know exactly which real coin serial number Alice later spends.

So withdrawal and spending become unlinkable.

That gives strong privacy.

9.4 What does Bob do?

Bob receives:

- SN
- $\text{sig}(SN)$

He can verify that the bank's signature is valid.

So Bob knows the coin was properly bank-signed.

9.5 What problem does this solve?

It solves the privacy issue of simple e-cash.

The lecture says the bank does not see the serial number, and privacy becomes comparable to physical paper money.

That is huge.

10) Chaumian E-Cash — pros and cons

The lecture summarizes:

Pros:

- digital
- payer and payee can be far apart
- peer-to-peer
- privacy maintained
- same privacy guarantee as physical paper money

Cons:

- bank is still the middle trusted party
- bank can censor
- double-spending detection/identification is possible only **after the fact**.

10.1 Why is Chaumian e-cash better than simple e-cash?

Because it fixes privacy.

The bank no longer sees everything about every coin's later use.

10.2 What is still wrong?

It is still centralized.

The bank is still necessary:

- to issue the coin
- to be trusted
- to potentially stop/censor activity

So privacy improved, but decentralization did not.

10.3 What about double-spending?

It is still not the perfect decentralized solution. Detection and identification happen only later, not by a global decentralized ledger in advance.

That is why Chaumian e-cash was an important step, but not the final answer.

11) Why didn't Chaumian E-Cash win commercially?

The lecture says Chaumian E-Cash did not succeed because:

- it was considered too private
- anonymous digital payments raised worries about tax evasion and money laundering
- online verification in the 1990s gave poor user experience
- business model gave banks little incentive
- result: too private for government, too inconvenient for users, too unattractive for banks.

This is a beautiful exam topic because it mixes technology and real-world adoption.

11.1 Too private for governments

Strong privacy can clash with state goals of surveillance, AML, KYC, and law enforcement.

11.2 Too inconvenient for users

In the 1990s, always-online verification was much less practical than today. Networks were weaker, slower, and less reliable.

11.3 Too unattractive for banks

Banks did not gain control from this system — they lost some of it.

So there was weak incentive for them to adopt it.

11.4 Final takeaway

A technology may be elegant, but still fail socially, politically, and commercially.

That is an important lesson.

12) Desirable features of money

The lecture lists:

- durable
- portable
- divisible
- fungible
- acceptable
- stable
- limited supply
- hard to counterfeit.

These are basic money qualities, and sir can easily ask them.

12.1 Durable

Money should last and not decay quickly.

12.2 Portable

It should be easy to carry or transfer.

12.3 Divisible

It should be splittable into smaller units.

12.4 Fungible

One unit should be interchangeable with another equal unit.

12.5 Acceptable

People should widely accept it in exchange.

12.6 Stable

Its value should not swing wildly in normal use.

12.7 Limited supply

It should not be easily inflated without control.

12.8 Hard to counterfeit

Forgery should be difficult.

The lecture asks whether these features matter for both digital and non-digital money. The answer is yes — though the way they are implemented differs.

13) Early digital currencies: shortcomings

The lecture then asks why earlier digital currencies failed, or in other words, what shortcomings Bitcoin addressed. It asks:

- Is it peer-to-peer?
- Does it avoid stealing?
- Is it widely accepted?
- Is there well-defined minting?
- Does it offer a ledger?
- Does it preserve privacy?
- Does it avoid double-spending?

This is basically the transition from older systems to Bitcoin.

Sir is showing that Bitcoin matters because it addresses multiple old weaknesses together.

14) Cryptocurrency challenges — and Bitcoin's answers

The lecture lists three major challenges:

- no stealing → solution: cryptographic digital signatures
- no double-spending → solution: global ledger
- minting → solution: proof of work.

This is one of the strongest compact explanation slides in the whole course.

14.1 No stealing

Only the rightful owner should move funds.

Solution: digital signatures.

14.2 No double-spending

The same money should not be spent twice.

Solution: a shared global ledger.

14.3 Fair minting

New money should not be created arbitrarily by random parties.

Solution: PoW-based block reward mechanism.

These three answer three central digital-currency problems.

15) CBDC, Bitcoin, and Chaumian e-cash — comparison

The lecture includes a comparison table. It says:

Chaumian e-cash:

- issuer is central bank or licensed bank
- strong privacy through blind signatures
- double-spending solved by bank database + cheater revealing
- centralized
- rejected because too private
- failed commercially

Bitcoin:

- no central issuer
- pseudonymous, not fully private
- double-spending solved by public blockchain consensus
- decentralized
- tolerated but regulated
- global adoption and ecosystem

CBDCs:

- central bank issued
- fully traceable
- central bank ledger
- centralized
- actively promoted
- pilots ongoing
- strong control but privacy concerns.

This table is extremely exam-friendly.

15.1 Chaumian e-cash

Best privacy, but centralized and politically problematic.

15.2 Bitcoin

Decentralized and censorship-resistant, but not perfectly private and still has volatility/scalability issues.

15.3 CBDC

Efficient and state-controlled, but weak on privacy.

This is an excellent “compare and contrast” question possibility.

Lecture 08 begins — deeper consensus and modern blockchain topics

16) Consensus in blockchains

The lecture defines consensus as reaching agreement in a group. It says blockchain needs consensus dynamically and on-the-fly, and it cannot take forever. Consensus is needed to avoid conflicting transactions like T1 and T2 from ending up in the same ledger if they appear in different journals. Dishonest nodes not reaching consensus is okay.

16.1 Why is “dynamic” important?

Because blockchain is a live system.

Transactions keep arriving continuously.

Consensus is not a one-time classroom vote. It is ongoing.

16.2 Why can't consensus take forever?

Because real payment systems need usable settlement time.

If agreement takes too long, the system becomes impractical.

16.3 Why is it needed?

Because without consensus, conflicting transactions could survive in different local views and corrupt the ledger.

Consensus prevents contradictory histories from both becoming accepted.

17) Consensus via majority

The lecture says that for many nodes, a unified correct ledger evolves if the **majority** is honest. But then it asks: what does majority mean? One user can create many dishonest nodes, so a **weighted majority** is considered — each node is assigned a weight.

This is extremely important.

17.1 Why not simple headcount?

Because one attacker could create many fake identities/nodes.

If every node counted equally, the attacker could cheat by Sybil-like multiplication of nodes.

17.2 What is weighted majority?

It means “majority” is measured by some costly or meaningful resource, not just raw number of machines/accounts.

That resource might be:

- hashing power in PoW
- stake in PoS

17.3 Why does this help?

Because it makes influence expensive.

You cannot cheaply dominate consensus just by spinning up many names.

18) Proof of Work consensus

The lecture says PoW has roots in spam filtering:

- use a moderately expensive but solvable puzzle,
- it does not hurt normal users much,

- but discourages spammers.

In blockchain:

- the first node to solve the computational puzzle wins,
- its result is easy for others to verify,
- and its block gets added.

18.1 Spam-filtering roots

The idea is simple:

make abusive behavior costly enough that attackers are discouraged.

18.2 In blockchain

The winning node demonstrates computational effort by solving the puzzle.

This is its proof of work.

18.3 Easy to verify, hard to produce

This is the key strength:

- honest network quickly checks the answer
- attacker had to spend real effort to find it

18.4 What if work were minimal?

The lecture asks this as a thought question.

If work is too easy, then block creation becomes cheap and manipulation becomes easier. So the security value of PoW would collapse.

19) Proof of Stake consensus

The lecture says PoS was designed to reduce high energy usage of PoW. Ethereum switched from PoW to PoS. In PoS:

- validators stake ETH
- the system randomly chooses a validator
- higher stake means higher chance
- no heavy computation is needed
- cheating causes loss of security deposit

- 51%+ of staked ETH would be needed to defraud the system.

19.1 Main difference from PoW

PoW ties influence to computational power.

PoS ties influence to staked economic value.

19.2 Why is PoS attractive?

Because it reduces energy cost.

19.3 Why is it still secure?

Because cheating becomes economically dangerous: the validator's own stake is at risk.

19.4 What does 51% mean here?

In PoS, the relevant majority is majority of stake, not majority of number of nodes.

20) Other consensus mechanisms

The lecture briefly lists:

- Proof of Activity
- Proof of Storage
- Proof of Authority, Burn, Capacity, Assignment, Transfer, Elapsed Time
- PBFT, which reaches consensus if more than 2/3 nodes are honest and is used when participants are known.

You probably do not need deep detail here yet, but you should understand the message:

consensus is a design space, not just PoW and PoS.

PBFT is especially important conceptually because it fits settings where membership is known, unlike Bitcoin-style open networks.

21) Smart contracts

Now the lecture moves to recent topics. It defines a smart contract as:

- a program that runs on the blockchain

- encodes a self-executing agreement based on predefined events
- automatically triggers settlement when conditions are met.

This is a major blockchain idea.

21.1 What is a smart contract really?

It is not “smart” because it thinks.

It is smart because it automatically executes predefined logic.

So if the conditions occur, the contract performs the programmed action.

21.2 Why is this useful?

Because it removes the need for a human intermediary to manually enforce the agreement every time.

21.3 Crop insurance example

The lecture gives crop insurance:

- farmers lock premium into the smart contract
- investors lock payout funds too
- if rainfall falls below a certain threshold, payout is triggered automatically.

This is a very good example because it shows automatic settlement based on an external condition.

21.4 Role of external data / oracle

The smart contract itself does not magically know rainfall. The example slide shows data coming from a weather API via the Chainlink decentralized oracle network. The diagram on page 41 visually shows farmer, investor, smart contract, weather API, and oracle connection.

That means:

blockchain code often needs trusted or semi-trusted external data feeds to react to real-world events.

22) Digital assets across multiple systems / tokenization

The lecture then discusses representing assets in new systems. Examples:

- wBTC for BTC
- USDC pegged to USD
- tokenized Apple stock
- tokenized Roosevelt Hotel
- tokenized gold.

The lecture says the purpose is that some things can be done better in the new system, such as:

- unlock huge value and use it programmatically
- make P2P transactions fast and cheap
- give access to people who otherwise do not have it
- allow fractional ownership
- make trading more convenient.

22.1 What is tokenization?

Tokenization means creating a digital token that represents value or ownership tied to some asset.

22.2 Why do it?

Because the blockchain environment may offer:

- programmability
- fractionalization
- easier transfer
- new market access
- direct trading

22.3 Examples

wBTC:

represents BTC in a form usable in another blockchain ecosystem.

USDC:

represents fiat-like dollar value in token form.

Tokenized stock:

lets equity-like exposure be traded as tokens.

Tokenized hotel / real estate:

allows fractional ownership and broader participation.

Tokenized gold:
makes gold exposure easier to transfer and trade digitally.

23) Tokenized stocks and cross-system value

The lecture asks whether stock exchanges already represent company value as stocks. Yes — but tokenization changes the mode of transfer. The slide says tokenization of stocks brings **P2P trading and transactions**, and asks whether the stock exchange can be bypassed.

23.1 What is new here?

Traditional stock exchanges already represent company value, but they do so through a conventional market infrastructure.

Tokenization adds:

- peer-to-peer transferability
- blockchain-based movement
- possibly global access and programmability

23.2 Example slide

The page 45 diagram shows a user buying \$50 worth of Apple stock in token form through a VASP, smart contract, stock broker, and safe custodian. The token is minted and payment released, and the user then can directly trade the tokenized asset with others.

So the system is not necessarily eliminating all traditional institutions — it often wraps them differently.

24) Layer 2

The lecture defines Layer 2 as a **secondary protocol built on top of an existing blockchain**. Its main goals are:

- higher transaction speeds
- possibly adding functionality the main chain does not offer

Examples:

- Stacks for Bitcoin

- Polygon for Ethereum.

24.1 Why Layer 2 exists

Main blockchains often face limits in:

- throughput
- speed
- cost
- feature flexibility

Layer 2 tries to solve these without fully replacing the main chain.

24.2 What is the basic idea?

Use the security or base existence of the main chain, but move some activity to a secondary structure that is faster or richer in features.

25) CBDC

Finally, the lecture defines CBDC as a **government-issued stablecoin** and notes:

- most fiat is already digital
- but CBDC differs because it is governed by the central bank, whereas ordinary digital fiat is handled by commercial banks.

25.1 What is a CBDC?

Central Bank Digital Currency.

It is state-controlled digital money directly under central bank governance.

25.2 Why is this different from digital bank money?

Because your current bank balance is usually a claim in the commercial banking system, not direct central-bank-issued retail money.

CBDC changes who governs the digital money structure.

25.3 Why link it with stablecoins?

Because stablecoins are token-like representations of fiat value, and the lecture frames CBDC as a government-issued version of that idea.

Week 4 — Full connected story

Now let me connect the whole week into one story.

Week 4 begins by revisiting money and asking whether fiat always works. Fiat money has no intrinsic value and is not backed by a valuable commodity; it works mainly because people trust the government and the monetary system. But when that trust breaks, inflation can spiral. The lecture uses Weimar Germany to show how war debt, reparations, loss of productive capacity, and massive money printing led to hyperinflation so severe that by November 1923 one US dollar was worth 4,210,500,000,000 German marks. The photos of gigantic denominations and wheelbarrows of money make the point visually: a currency can become functionally useless if trust and discipline collapse.

The lecture then asks what a good digital currency should achieve: no stealing, no double-spending, fair minting, peer-to-peer payments, no always-online intermediary, and privacy like cash. Traditional ePayments do not satisfy this fully because banks remain in the middle: they can fail, censor, delay, and observe everything. Early digital cash systems tried to improve this. Simple e-cash made payments digital and P2P-like, but suffered from double-spending and privacy problems because serial numbers let the bank trace activity. Chaumian e-cash improved privacy dramatically using blinded signatures, so the bank could authorize coins without seeing their true serial numbers. But it still depended on a central bank-like issuer, could still be censored, and did not win commercially because it was too private for governments, inconvenient for users, and unattractive for banks.

This leads directly to why Bitcoin matters. Earlier digital currencies lacked a decentralized solution to stealing, double-spending, and fair minting. Bitcoin answers these with digital signatures, a global ledger, and Proof of Work. Lecture 08 then returns to consensus: blockchain must dynamically reach agreement so conflicting transactions in different journals do not both become part of the same ledger. A simple count of nodes is not enough because one attacker can create many fake identities, so weighted majority is used — hashing power in PoW or stake in PoS. PoW makes influence costly through computational work; PoS makes it costly through locked economic stake. Other mechanisms exist too, including PBFT for known-participant settings.

The lecture ends by widening the scope of blockchain beyond currency. Smart contracts are programs on the blockchain that automatically execute agreements when predefined conditions are met, as shown by the crop insurance example using oracle-fed rainfall data. Tokenization shows how assets like BTC, dollars, stocks, real estate, and gold can be represented in new systems to gain programmability, P2P transfer, broader access, and fractional ownership. Layer 2 protocols aim to increase speed and add functionality above the base chain. CBDCs represent a state-controlled digital-money direction, where the central bank directly governs

digital currency rather than leaving digital balances mainly to commercial banks. So Week 4 is really the bridge between “why older money systems fail” and “why blockchain became a new design space.”

Most important Week 4 topics to memorize

These are the high-priority topics:

- why fiat depends on trust
 - hyperinflation of Germany 1923
 - digital currency requirements
 - traditional ePayments pros and cons
 - why banks being “in the way” matters
 - simple e-cash and double-spending
 - serial numbers in simple e-cash
 - privacy problem in simple e-cash
 - Chaumian e-cash and blinded signatures
 - why Chaumian e-cash still failed
 - desirable features of money
 - what Bitcoin solved from earlier digital currencies
 - consensus in blockchains
 - weighted majority
 - PoW vs PoS in Week 4 framing
 - smart contracts
 - oracle / external data
 - tokenization
 - Layer 2
 - CBDC vs digital fiat
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Ultra-short Week 4 revision

Fiat money works only as long as people trust the government and monetary system; if trust collapses, hyperinflation can follow, as shown by Weimar Germany where the mark became almost worthless. Digital money should prevent stealing and double-spending, support peer-to-peer payments, avoid dependence on always-online intermediaries, and preserve privacy. Traditional ePayments are digital but not peer-to-peer because banks remain central and can fail, censor, delay, and monitor users. Simple e-cash makes payments digital and somewhat P2P, but double-spending and privacy remain problems. Chaumian e-cash uses

blinded signatures to protect privacy, but still depends on a trusted bank and failed commercially. Bitcoin's big contribution is combining digital signatures, a global ledger, and Proof of Work to address stealing, double-spending, and fair minting in a decentralized way. Consensus must dynamically stop conflicting transactions from entering the same ledger, so weighted majority is used in systems like PoW and PoS. Smart contracts enable automatic on-chain execution of agreements, tokenization represents assets in new programmable systems, Layer 2 improves speed/functionality above main chains, and CBDCs are central-bank-governed digital money.